

```
mysql> SELECT name, salary, bonus, (salary + bonus) AS total_income FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | total_income |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 135000       |
| Larry Page | 110000 | 12000 | 122000       |
| Sundar Pichai | 100000 | 10000 | 110000       |
| Satya Nadella | 95000  | 9000  | 104000       |
| Mark Zuckerberg | 115000 | 13000 | 128000       |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name, salary, bonus, (salary - bonus) AS after_deduction FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | after_deduction |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 105000          |
| Larry Page | 110000 | 12000 | 98000           |
| Sundar Pichai | 100000 | 10000 | 90000           |
| Satya Nadella | 95000  | 9000  | 86000           |
| Mark Zuckerberg | 115000 | 13000 | 102000          |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name, salary, (salary * 2) AS double_salary FROM employee;
+-----+-----+-----+
| name      | salary | double_salary |
+-----+-----+-----+
| Elon Musk | 120000 | 240000        |
| Larry Page | 110000 | 220000        |
| Sundar Pichai | 100000 | 200000        |
| Satya Nadella | 95000  | 190000        |
| Mark Zuckerberg | 115000 | 230000        |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name, salary, (salary / 10) AS salary_div_10 FROM employee;
+-----+-----+-----+
| name      | salary | salary_div_10 |
+-----+-----+-----+
| Elon Musk | 120000 | 12000.0000    |
| Larry Page | 110000 | 11000.0000    |
| Sundar Pichai | 100000 | 10000.0000    |
| Satya Nadella | 95000  | 9500.0000     |
| Mark Zuckerberg | 115000 | 11500.0000    |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name, salary, bonus, (salary % bonus) AS remainder FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | remainder |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 0          |
| Larry Page | 110000 | 12000 | 2000      |
| Sundar Pichai | 100000 | 10000 | 0          |
| Satya Nadella | 95000  | 9000  | 5000      |
| Mark Zuckerberg | 115000 | 13000 | 11000     |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary = 100000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 3       | Sundar Pichai | 100000 | 10000 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary <> 100000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 1       | Elon Musk | 120000 | 15000 |
| 2       | Larry Page | 110000 | 12000 |
| 4       | Satya Nadella | 95000  | 9000  |
| 5       | Mark Zuckerberg | 115000 | 13000 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary > 100000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 1       | Elon Musk | 120000 | 15000 |
| 2       | Larry Page | 110000 | 12000 |
| 5       | Mark Zuckerberg | 115000 | 13000 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary < 100000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 4       | Satya Nadella | 95000  | 9000  |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary >= 110000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 1       | Elon Musk | 120000 | 15000 |
| 2       | Larry Page | 110000 | 12000 |
| 5       | Mark Zuckerberg | 115000 | 13000 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE salary <= 95000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 4       | Satya Nadella | 95000  | 9000  |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT name, salary, bonus, (salary & bonus) AS bitwise_and FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | bitwise_and |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 4224        |
| Larry Page | 110000 | 12000 | 11424       |
| Sundar Pichai | 100000 | 10000 | 1536       |
| Satya Nadella | 95000  | 9000  | 8968       |
| Mark Zuckerberg | 115000 | 13000 | 8           |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name, salary, bonus, (salary | bonus) AS bitwise_or FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | bitwise_or |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 130776     |
| Larry Page | 110000 | 12000 | 110576     |
| Sundar Pichai | 100000 | 10000 | 108464     |
| Satya Nadella | 95000  | 9000  | 95032      |
| Mark Zuckerberg | 115000 | 13000 | 127992     |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name, salary, bonus, (salary ^ bonus) AS bitwise_xor FROM employee;
+-----+-----+-----+-----+
| name      | salary | bonus | bitwise_xor |
+-----+-----+-----+-----+
| Elon Musk | 120000 | 15000 | 126552     |
| Larry Page | 110000 | 12000 | 99152      |
| Sundar Pichai | 100000 | 10000 | 106928     |
| Satya Nadella | 95000  | 9000  | 86064      |
| Mark Zuckerberg | 115000 | 13000 | 127984     |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name, salary, (~salary) AS bitwise_not FROM employee;
+-----+-----+-----+
| name      | salary | bitwise_not |
+-----+-----+-----+
| Elon Musk | 120000 | 18446744073709431615 |
| Larry Page | 110000 | 18446744073709441615 |
| Sundar Pichai | 100000 | 18446744073709451615 |
| Satya Nadella | 95000  | 18446744073709456615 |
| Mark Zuckerberg | 115000 | 18446744073709436615 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name, salary, (salary < 1) AS left_shift FROM employee;
+-----+-----+-----+
| name      | salary | left_shift |
+-----+-----+-----+
| Elon Musk | 120000 | 240000     |
| Larry Page | 110000 | 220000     |
| Sundar Pichai | 100000 | 200000     |
| Satya Nadella | 95000  | 190000     |
| Mark Zuckerberg | 115000 | 230000     |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name, salary, (salary > 1) AS right_shift FROM employee;
+-----+-----+-----+
| name      | salary | right_shift |
+-----+-----+-----+
| Elon Musk | 120000 | 60000       |
| Larry Page | 110000 | 55000       |
| Sundar Pichai | 100000 | 50000       |
| Satya Nadella | 95000  | 47500       |
| Mark Zuckerberg | 115000 | 57500       |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM employee WHERE salary > 100000 AND bonus > 12000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 1      | Elon Musk | 120000 | 15000 |
| 5      | Mark Zuckerberg | 115000 | 13000 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE salary < 100000 OR bonus < 10000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 4      | Satya Nadella | 95000  | 9000  |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM employee WHERE NOT salary = 95000;
+-----+-----+-----+
| emp_id | name      | salary | bonus |
+-----+-----+-----+
| 1      | Elon Musk | 120000 | 15000 |
| 2      | Larry Page | 110000 | 12000 |
| 3      | Sundar Pichai | 100000 | 10000 |
| 5      | Mark Zuckerberg | 115000 | 13000 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM employee WHERE salary BETWEEN 100000 AND 120000;
```

emp_id	name	salary	bonus
2	Larry Page	107000	12000

1 row in set (0.00 sec)

```
mysql> SELECT * FROM employee WHERE name IN ('Elon Musk', 'Mark Zuckerberg');
```

emp_id	name	salary	bonus
1	Elon Musk	125000	15000
5	Mark Zuckerberg	5000	13000

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM employee WHERE name LIKE '%%';
```

emp_id	name	salary	bonus
3	Sundar Pichai	200000	10000
4	Satya Nadella	47500	9000

2 rows in set (0.00 sec)

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM employee WHERE bonus IS NULL;
Empty set (0.00 sec)
```

```
mysql> SELECT * FROM employee WHERE bonus IS NOT NULL;
```

emp_id	name	salary	bonus
1	Elon Musk	125000	15000
2	Larry Page	107000	12000
3	Sundar Pichai	200000	10000
4	Satya Nadella	47500	9000
5	Mark Zuckerberg	5000	13000

5 rows in set (0.00 sec)

5 rows in set (0.00 sec)

```
mysql>
mysql> SELECT name, salary
-> FROM employee e
```

```
-> WHERE EXISTS (
-> SELECT 1 FROM employee
-> WHERE bonus > 12000 AND emp_id = e.emp_id
-> );
```

name	salary
Elon Musk	125000
Mark Zuckerberg	5000

2 rows in set (0.00 sec)

2 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM employee WHERE salary > ANY (SELECT bonus FROM employee);
```

emp_id	name	salary	bonus
1	Elon Musk	125000	15000
2	Larry Page	107000	12000
3	Sundar Pichai	200000	10000
4	Satya Nadella	47500	9000

4 rows in set (0.00 sec)

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM employee WHERE salary > ALL (SELECT bonus FROM employee);
```

emp_id	name	salary	bonus
1	Elon Musk	125000	15000
2	Larry Page	107000	12000
3	Sundar Pichai	200000	10000
4	Satya Nadella	47500	9000

4 rows in set (0.00 sec)

4 rows in set (0.00 sec)

```
mysql> UPDATE employee SET salary = salary + 5000 WHERE name = 'Elon Musk';
Query OK, 1 row affected (0.02 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE employee SET salary = salary - 3000 WHERE name = 'Larry Page';
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE employee SET salary = salary * 2 WHERE name = 'Sundar Pichai';
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE employee SET salary = salary / 2 WHERE name = 'Satya Nadella';
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE employee SET salary = salary % 10000 WHERE name = 'Mark Zuckerberg';
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT * FROM employee;
```

emp_id	name	salary	bonus
1	Elon Musk	125000	15000
2	Larry Page	107000	12000
3	Sundar Pichai	200000	10000
4	Satya Nadella	47500	9000
5	Mark Zuckerberg	5000	13000

5 rows in set (0.00 sec)